

Electrical & Electronic Engineering (EEE) and Computer Engineering (CE) Final Year Projects (2026-2027)

Project Title: Physics-informed neural networks for hybrid dynamics with LLM-assisted experimental design

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Classification: Control, Software, Intelligent Systems.

What are physics informed neural networks (PINNs)? PINNs combine data-driven learning with known physical laws. Rather than fitting measurements alone, their training objective can also penalise violations of algebraic equations, ODEs or PDEs. This can improve physical consistency, but selecting an effective formulation and loss function remains challenging, especially for non-smooth systems.

What are hybrid systems? Hybrid dynamical systems combine continuous-time behaviour with discrete switching or non-smooth effects. Examples in robotics include contact, impacts and Coulomb friction. Such behaviour can be represented in several ways, including flow/jump models, complementarity conditions and differential inclusions; recent formulations suggest that the latter two may be particularly useful for PINNs.

Objective and methodology: In this project you will develop and study PINN methodologies for hybrid dynamical systems. The work will begin with simple systems in order to build intuition and will then investigate formulations suitable for switching and non-smooth dynamics. Once a reliable PINN baseline has been established, the project will explore whether large language models can assist selected parts of the modelling workflow, particularly experimental design and model reparameterisation.

What it involves: Your work will combine mathematical modelling, simulations and machine learning. Familiarity with calculus and programming, esp. Python, will be very helpful.

This project will require completion of a number of tasks including but not exclusively the following:

Expected Tasks

1. Extensive literature review, esp. on PINNs, hybrid dynamical systems and recent approaches for non-smooth dynamics.
2. Formulation of several PINN estimation problems for simple and hybrid dynamical systems.
3. Implementation of PINN estimation pipelines with different hybrid-system representations and loss functions; appropriate validation/testing and reporting of the results.
4. Experimentation with different input signals to create informative datasets, including investigation of optimal experimental design and possible LLM-assisted approaches.
5. Application of the methodology to robotic systems, with possible investigation of LLM-assisted model reparameterisation or experimental design where appropriate.

Learning Outcomes

Upon completion of the project, you will expect to have:

6. Solid understanding of various types of PINNs and their application to hybrid systems.
7. The ability to design suitable physics-informed loss functions and use data to train PINNs in Python or another programming language.
8. Solid understanding of hybrid system dynamics and different mathematical representations of switching and non-smooth behaviour.
9. Ability to design informative simulation experiments and critically assess where LLM-based methods may add value.

If conducted well, it is likely that the results of this project can lead to a publication.

MEng Project

If assigned as an MEng EEE/CE project, the scope of work will be extended and include additional tasks and learning outcomes. MEng extensions must be discussed and agreed with the supervisor and moderator.

References

- [1] P. Sopasakis, Physics informed neural networks (PINNs); Python notebook, available on GitHub, 2026
- [2] S.L. Brunton and J.N. Kutz, Data-Driven Science and Engineering, Cambridge University Press, 2019
- [3] R. Goebel, R.G. Sanfelice, and A.R. Teel, Hybrid dynamical systems: modelling, stability, and robustness, Princeton University Press, 2012.
- [4] R. Yao, A. McAleenan, Y. Jian, M. Van, J. Moore and Y. Jin, "A Novel Dynamic Modeling Method of a New 5-Bar Spherical Parallel Robot for Ophthalmic Surgery via Screw Theory," 2025 IEEE International Conference on Robotics and Biomimetics (ROBIO), Chengdu, China, 2025, pp. 466-471